1 After taking measurements of the quantities in the expression $\frac{3PQ^4}{R^2}$, the total uncertainty is calculated as 11%.

Which individual percentage uncertainties of P, Q and R when combined give this total of 11%?

	P / %	*Q* / %	*R* / %	z / %
A	1.0	1.0	2.0	**9.0**
B	2.0	2.0	1.5	**13**
C	3.0	1.0	2.0	**11**
D	5.0	2.0	1.0	**15**

Ans: C

Let $z = \frac{3PQ^4}{R^2}$

$$\frac{\Delta z}{z} = \frac{\Delta P}{P} + 4\frac{\Delta Q}{Q} + 2\frac{\Delta R}{R}$$

C is the only option which gives 11%